

# KAHBOD AEINI

Computer Science Department, University of Calgary, Calgary, Alberta, Canada

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## EDUCATION

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- University of Waterloo** Starting in Jan 2026  
PhD in Electrical and Computer Engineering  
Supervisor: Prof. Wojciech Golab
- University of Calgary** 2023 - Ongoing  
Master of Computer Science  
Supervisor: Prof. Philipp Woelfel  
Thesis: *Space-Optimal Randomized Wait-Free Lock*
- Sharif University of Technology** 2019 - 2023  
Bachelor of Computer Engineering  
Thesis: *Fraud Detection from Sequences of Bank Transactions Using Various Machine Learning Algorithms*
- National Organization for Development of Exceptional Talents** 2013 - 2019  
High School Diploma in Mathematics and Physics

## RESEARCH INTERESTS

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Distributed Systems, Concurrent & Randomized Algorithms, Advanced Data Structures, Blockchain, Database Systems, Machine Learning and Computational Biology

## PUBLICATIONS

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**Kahbod Aeini**, Dante Bencivenga, George Giakkoupis, Philipp Woelfel.  
*Simple and Efficient Randomized Wait-Free Lock*.  
Submitted to ACM-SIAM Symposium on Discrete Algorithms (SODA 2026), under review.

## RESEARCH EXPERIENCE

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- Distributed Algorithms**, Research Assistant, Graduate Student Jun 2024 - Ongoing  
Supervisor: Prof. Philipp Woelfel, University of Calgary  
Designed and analyzed randomized wait-free lock algorithms achieving optimal trade-offs between space and time complexity in shared memory systems, including the first space-optimal algorithm and complementary time-optimal variants.
- Bioinformatics**, Research Assistant (Volunteer, Remote) Feb 2023 - Aug 2023  
Supervisor: Prof. Mehdi Pirooznia, Johns Hopkins University  
Conducted feature selection and developed machine learning models to predict immune cell composition in lung cancer. Performed identification of immune cell types, analysis of differentially expressed genes, functional enrichment analysis, and classification of cancer subtypes using machine learning techniques.
- Machine Learning**, Research Assistant Aug 2022 - Nov 2022  
Supervisor: Prof. Abbas Heydarnoori, Intelligent Software Engineering (ISE) Lab  
Developed deep learning models for code prediction and semantic analysis of Python programs using advanced feature extraction from Jupyter notebooks in the KGTorrent dataset, contributing to automated programming assistance and code comprehension systems.
- Deep Learning**, Research Assistant, Machine Learning Oct 2021 - Feb 2022  
Supervisor: Dr. Fatemeh Baharifar & Dr. Vahid Motaghed, *IPM*

Developed and evaluated face clustering algorithms using graph convolutional networks (GCN) for linkage prediction and pairwise classification approaches.

## TEACHING EXPERIENCE

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### University of Calgary

2023 - 2025

Led and managed TA teams across undergraduate and graduate courses (up to 300 students), covering algorithms, programming, and theoretical computer science.

- **Programming Paradigms (Fall 2025)** — Managed 6 TAs, organized and held tutorials in Haskell & Prolog for 75+ students.
- **Randomized Algorithms (Graduate, Fall 2024)** — Delivered tutorials on randomized data structures, lower bound techniques, and complexity classes.
- **Theoretical Foundations of Computer Science II (Winter 2024 & 2025)** — Coordinated 7 TAs, supported 150+ students in automata theory and probability.
- **Python Programming (Fall 2023)** — Taught fundamentals of Python, data structures, and algorithms for 50+ students.

### Sharif University of Technology

2020 - 2022

Served as Head TA across multiple undergraduate courses, leading and mentoring TA teams of 10-25 members while supporting 200+ students in core computer science subjects.

- **Data Structures and Algorithms** — Designed assignments & projects, led TA group of 25+.
- **Numerical Calculations** — Developed final project curriculum, led TA group of 10+.
- **Mobile Programming** — Designed and graded Android development projects.

## WORK EXPERIENCE

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### Data Engineer, AI Lida

Feb 2022 - Sep 2022

- Developed, maintained and optimized data pipelines for ETL processes using Python and PostgreSQL.
- Collaborated with the security department to gather requirements and deliver data solutions that supported business objectives.
- Interviewed, hired, trained and supervised a team of 5 data engineer interns, providing mentorship and performance feedback.

## AWARDS AND SCHOLARSHIPS

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### Departmental Research Award, Department of Computer Science, University of Calgary

2023

Won the RA award, which is granted to the best students in each department to step away from TA responsibilities and focus on their research

### Top 100 Students in Country

2019

Ranked 94 (31st in the Region) in National Universities Entrance Exam among around 160,000 participants

### National Elites Fellowship

2019

Fellowship in Iran's National Elites Foundation

## SKILLS

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**Languages:** Python, R, Haskell, Java, C/C++, Prolog, Android

**Frameworks & Tools:** PyTorch, TensorFlow, scikit-learn, Django, Numpy, Pandas, OpenCV, limma

**Databases:** PostgreSQL, MongoDB, MySQL

**Other:** Linux, NixOS, Kali